

Intra-Aortic Balloon Pump/Impella Device

Guideline Title: Intra-Aortic Balloon Pump/Impella Device

Guideline Number: GUID-3203-PR023

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 6/8/2021	Developed by: Air Care Team
Approved by: Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

CCCXXVI. **PURPOSE:** Intra-aortic balloon pump (IABP) therapy is designed to increase coronary artery perfusion, increases systemic perfusion, decrease myocardial workload, and decrease afterload. It is an acute, short-term therapy for patients with reversible left ventricular failure.

CCCXXVII. **DEFINITIONS:**

- A. IABP – Intra-aortic Balloon Pump
- B. CXR – Chest X-ray

CCCXXVIII. **DEPARTMENT GUIDELINE:**

A. **EQUIPMENT:**

- i. Transport Intra-aortic Balloon Pump
 - 1. Our pump currently is the Cardiosave IABP Rescue produced by Maquet, which is the smallest and lightest model. Weight: 24.1kg/53lb, Size in cart: 30.7inches high x 16 inches deep x 13 inches wide.
- ii. Spare battery and electrical cord
- iii. 250 ml or > of NS
- iv. Pressure Bag
- v. Invasive monitoring tubing
- vi. 3 way stop cock
- vii. 60ml catheter tip syringe

B. **INDICATIONS:**

- i. Decompensated systolic heart failure
- ii. Acute Myocardial Infarction where cardiogenic shock cannot be reversed
- iii. Acute mitral regurgitation

C. **CONTRAINDICATIONS:**

- i. Moderate to severe aortic insufficiency
- ii. Thoracic and abdominal aortic aneurysms

D. **PROCEDURE:**

- i. Establish care defined by *GUID-G002 – General Patient Care*.
- ii. Obtain the following information if available from the staff or from the patient's record: IABP insertion site, method of visualization of correct placement and full inflation of the balloon via fluoroscopy or viewing of the CXR, Catheter type and brand, size, balloon volume, technique of insertion (sheath or sheath-less), size and location of sheath, and indication for placement.
 1. If the patient is connected to an IABP from a non-Orlando Health facility, ensure that all connectors are present. Contact sending facility cath lab for adapters.
- iii. Complete an assessment noting specifically
 1. The patient:
 - a. LOC: thrombi may develop and dislodge
 - b. Left arm: Skin color, temperature. Cap refill, quality of radial pulse and compare to right side. Place pulse ox on left hand if possible. (If balloon migrates upwards, occluding the left subclavian, you will have decreased perfusion to the arm)
 - c. Abdomen: swelling, pain, nausea, decreased/absent bowel sounds. (May indicate a balloon migration and an occlusion of the renal, celiac, or mesenteric arteries)
 - d. Leg or arm with insertion site: Skin color, temperature, cap refill, compare skin and pulses to opposite extremity. Immobilize the leg with a knee immobilizer or soft restraint if needed to prevent patient from bending of the knee and kinking of the IABP catheter
 - e. Insertion site: location, condition, dressing- intact, presence of bleeding, or hematoma. Mark the position of the catheter at the insertion site before moving the patient
 - f. Urine output: Foley draining should be $>0.5\text{ml/kg/hr}$.
 2. Review Labs: Complete blood count (CBC), platelet, and coagulation tests.
 3. Review pulmonary artery pressures (PA), Cardiac output (CO) or index measurements (CI) if available.
 4. The waveform
 - a. Place the pump in 1:2 mode if patient can tolerate. Print a strip of the waveform. (Pt should augment at equal to or greater than the unassisted systole.)

- b. Balloon augmentation should occur after the aortic valve closes. A sharp V wave should be noted, and the diacritic notch disappears.
 - c. Balloon assisted aortic end diastolic pressure should be lower than patient's unassisted diastolic pressure.
 - d. The assisted systolic pressure will be less than the unassisted systolic pressure.
 - e. Balloon pressure waveform should have the following: Filling pressure above zero, sharp up stroke, pressure plateau, rapid deflation
- 5. Review all IV fluid and medication infusions and transfer medications to transport pumps
- iv. Set up the transport IABP
 - 1. Check arterial line and other transducers for compatibility and change if needed.
 - 2. Check for fiber-optic sensor connector.
 - 3. Plug IABP model to AC power. Turn on pump and allow system to perform a self-check. Check helium level and battery levels.
 - 4. Place ECG skin leads on patient and monitor ECG on IABP screen.
 - 5. Pressure transducer: Open transducer vent to atmosphere. Press Zero pressure key. Close pressure transducer vent to atmosphere and assess pressure wave on display screen.
 - 6. Transfer the balloon lumen tubing to the IABP console.
 - 7. Check settings on IABP to match previous settings. Timing should be "auto". Place pump initially in 1:2 mode and print waveform strip to ensure accurate timing and augmentation.
 - 8. Start pumping by removing it from standby mode.
- v. Prepare patient for transport and move patient over to stretcher. Reassess patient and insertion site. Keep patient in supine position with HOB elevated no more than 30 degrees.
- vi. Load patient and IABP. Secure pump with two approved straps. Use caution when loading. If IABP disconnects, immediately reattach the patient's balloon to the chamber.
- vii. Reassess patient and monitor IABP tubing for connectivity, blood, possible dislodgment. Complete a cardiovascular and peripheral reassessment every 15 minutes. Monitor for signs of inappropriate IABP placement or balloon perforation.

CCCXXIX. TROUBLE SHOOTING

- A. If cardiac rhythm changes to asystole, change trigger to atrial pressure.
Decrease augmentation to 50%.
- B. When cardioverting/defibrillating, stand clear from patient and equipment.
- C. Reasons pump is not augmenting
 - i. Tachycardia or Bradycardia: affects cardiac output
 - ii. Hypovolemia: MAP<50 at the time of insertion is not enough to augment
 - iii. SVR>1200: Post coronary artery bypass grafting (CABG) or on high dose vasopressors, pump is unable to compete with increased afterload.
 - iv. Balloon volume is too low.
- D. Timing Errors
 - i. Early inflation: IABP prior to aortic valve closure (dicrotic notch)
 - 1. Will increase left ventricular wall stress, afterload, and MVO₂
 - ii. Late inflation: Inflation markedly after closure of aortic valve
 - 1. Produces suboptimal coronary perfusion
 - iii. Early deflation: Sharp drop following diastolic augmentation
 - 1. Causes increased assisted systolic pressure, suboptimal coronary flow, and suboptimal afterload reduction
 - iv. Late deflation: Afterload reduction is absent
 - 1. Left ventricle is ejecting against a greater resistance
- E. Loss of Vacuum of IABP Failure
 - i. Leak in the gas: check and tighten connections on the pneumatic tubing
 - ii. Check the power source
 - iii. Hand inflate and deflate the balloon every 5 minutes with half the total balloon volume
- F. Suspected Balloon Perforation
 - i. Observe for a loss of >10mm in diastolic augmentation
 - ii. Check for blood in the catheter
 - iii. Assess for changes in the balloon pressure waveform
 - iv. Put IABP in standby
 - v. Clamp the IABP catheter to prevent arterial blood back up
 - vi. Disconnect IABP catheter from the IABP console
 - vii. Call Control to notify the physician – The catheter should not lie dormant for longer than 30 minutes
- G. Gas loss: Pressure waveform falls below baseline
- H. Kink: Rounded balloon waveform
- I. Poor cardiac function or low vascular resistance: Refer to medical management per *GUID-3203-M011 Hypotension*.

CCCCXX. **DOCUMENTATION:**

- A. EMS Charts

CCCCXXI. **REFERENCES:**

- A. [Www.getinge.com](http://www.getinge.com) (Previously Maquet)